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Modeling a Knowledge Graph for Vulnerability and Real-World Exploitation Analysis

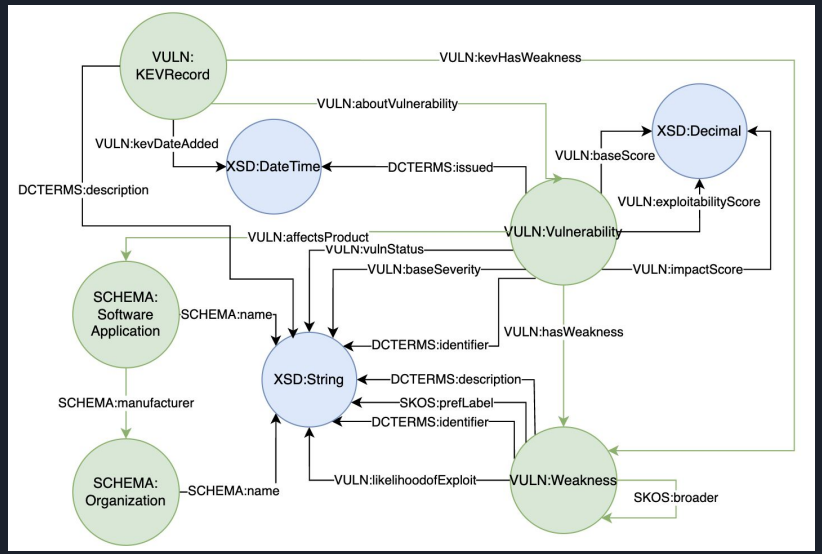
Software vulnerabilities are a major cybersecurity risk
Thousands of vulnerabilities are disclosed each year in the National Vulnerability Database (NVD) but only a small subset becomes actively exploited in real-world attacks (KEV).

Problem
Existing datasets (NVD, CWE, KEV) are:

- structured but not integrated for analysis;
- do not support cross-vendor or weakness-level exploration.

Goal
Build a Knowledge Graph integrating vulnerabilities, weaknesses, products and exploited vulnerabilities to enable structured and reproducible analysis of vulnerability patterns.

Approach



Data sources:

- NVD vulnerability dataset (CVE).
- MITRE Common Weakness Enumeration (CWE).
- CISA Known Exploited Vulnerabilities (KEV).

Data integration pipeline:

1. Collecting CVE, CWE, KEV datasets.
2. JSON/XML ⟶ CSV transformation.
3. Building RDF knowledge graph.
4. SHACL Validation.
5. SPARQL Analysis.

Results

Query 1: Product most affected by exploited vulnerabilities.

- Analysis of KEV-linked vulnerabilities shows which software product is most frequently targeted in real-world exploits.

productName	nExploitedVulns
Windows Server 2022 23h2	25

Query 2: Inconsistencies between CVE and KEV weakness annotations.

- Comparing NVD and KEV data reveals cases where exploited vulnerabilities are linked to different CWE categories.

cveId	nvdCWE	kevCWE
CVE-2025-32706	CWE-20	CWE-122

Insights, Limitations and Future work

Insights

- Knowledge graphs enable structured cybersecurity analysis.
- Integrating CVE, CWE and KEV reveals relationships not visible in isolated datasets.
- The graph supports reproducible cybersecurity analysis through SPARQL queries.

Limitations

- KEV includes only partial ground truth for exploitation.
- Vulnerability datasets are incomplete.

Future work

- Graph Neural Networks to predict which vulnerabilities are likely to be exploited.